

Assignment 4: Diffusion Models*March 2026**March 2026*

The purpose of this assignment is to give you hands-on experience designing and training of Diffusion Models.

Due Data: The assignment is due April 3rd, 2026 by 11:59 pm (electronic submission through UMLearn).

Assignments should be performed individually, but feel free to interact with your classmates. Plagiarism will not be tolerated. All codes should be well commented and any reference material used should be cited appropriately. If you use ChatGPT (or similar) you must include your prompts and responses as an appendix, along with an explanation of how you used the responses to guide your work and/or report writing.

Important: Please include the following signed statement with your submission.

I, [insert name] attest that the work I am submitting is my own work and that it has not been copied/plagiarized from online or other sources. Any sourced material used for completing this work has been properly cited. [Signature]

Introduction

You have been supplied with the Jupyter notebooks related to Denoising Diffusion Probabilistic Model (DDPM) taken from <https://keras.io/examples/generative/ddpm/#introduction>

Problem 1 : Comparing DDPM and WGAN-GP

For this problem your goal is to assess and evaluate the relative strengths and weaknesses of a denoising diffusion probabilistic model (DDPM) compared to WGAN-GP. For a fair comparison, use the same dataset you applied in Assignment 3, Problem 1, Part A for both models. Your analysis will be based on a comparison of the generated images, as well as the computational requirements to train the two models.

- Using the provided Jupyter notebooks, implement a diffusion model for the dataset you used in your WGAN-GP assignment.
- Train both models using the same set of hyperparameters where applicable. Note that some model-specific parameters will differ (e.g., noise schedule for the diffusion model and gradient penalty for WGAN-GP). Make reasoned choices for these parameters based on best practices or preliminary experiments. Be sure to provide a summary and justification for the choices of the relevant model parameters.
- Evaluate both models using qualitative and quantitative metrics. Consider using Inception Score (IS) or Fréchet Inception Distance (FID) for quantitative evaluation and include visual samples from both models for qualitative assessment.
- **(Graduate students only)** Compare the training dynamics of both models. Provide training curve plots and discuss any significant observations regarding convergence rates, stability, and overall behavior.
- Discuss the pros and cons of each model based on your experiments. Consider aspects such as training stability, complexity, quality, and diversity of the generated samples.

Discussion Question

(Graduate students only) Assuming that training went well for both models, can you provide some justification arguing for the robustness of diffusion models over GANs? For example, can you demonstrate a sensitivity to training parameters for WGAN-GP that doesn't exist for the DDPM?

Hand-in Report

Submit a report summarizing the relevant answers to Problem 1 and the associated discussion question. Include code snippets and explanations that facilitate our understanding of your solutions and allow us to interpret your results without resorting to running your code.

A Jupyter Notebook that is easy to navigate, is pre-run, and has ample markdown answering questions is sufficient.

When you submit your report, also submit your code in the event that we need to run it to validate your claims.